

# Pension Run

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## 1. Introduction

Why do retirees give up longevity insurance? In the canonical life-cycle model, an individual who faces an uncertain lifetime should convert a substantial share of wealth into an annuity. Pooling mortality risk allows an annuity to sustain consumption for as long as the retiree lives, a protection that self-insurance cannot provide as cheaply ([Davidoff, Brown, & Diamond, 2005](#); [Yaari, 1965](#)). Yet observed annuitization is limited: even when retirees are offered a direct choice between a lump-sum benefit and a stream of lifetime payments, many choose the lump sum. This gap between the theoretical value of longevity insurance and actual choices is the annuitization puzzle ([Benartzi, Previtro, & Thaler, 2011](#); [Brown, Kling, Mullainathan, & Wrobel, 2008](#)).

Most explanations of the puzzle treat the value of the promised annuity as independent of other retirees' choices. That assumption is natural for an actuarially fair contract backed by a solvent insurer, but it is less innocuous for benefits paid by a financially vulnerable pension fund. A lifetime payment is valuable only while the provider remains able and willing to pay. When a fund is expected to become depleted, the effective maturity of its monthly benefit is shortened. A retiree may then rationally prefer an immediate lump sum even when the scheduled, undiscounted value of monthly benefits is larger. Low annuitization can therefore reflect not only preferences for liquidity or bequests, but also exposure to provider-solvency risk.

The central tension is that the annuity is the more generous option if all promised payments are honored, yet its generosity can undermine the promise itself. When early cohorts choose monthly benefits, they build a persistent stock of long-lived liabilities. In the environment we study, those obligations eventually deplete the fund, whereas the fund remains sustainable if every generation instead takes the less generous lump-sum benefit. Insolvency is therefore not simply an exogenous reason to avoid annuitization; it can be the equilibrium consequence of earlier cohorts accepting the annuity offer.

This intergenerational feedback gives the pension run two stages. Early recipients choose the more valuable monthly benefit and gradually weaken the fund's solvency. Once the accumulated obligations make depletion sufficiently likely, later retirees discount the promised

stream and switch toward immediate lump-sum claims. These large current withdrawals can bring depletion forward and further reduce the value of annuitization. We call this feedback a *pension run*. Unlike a standard bank run, retirees do not withdraw identical demandable claims at the same time. They make a one-time choice as they retire, so aggregate fragility develops through a sequence of benefit-maturity decisions across generations.

This paper develops a continuous-time model of that mechanism. New retirees choose between an immediate lump-sum payment and a life-contingent monthly benefit. Their choice depends on the present value of monthly payments, which in turn depends on the stochastic time at which the pension fund is depleted. Individual choices determine both current withdrawals and the stock of monthly beneficiaries, feeding back into the fund's dynamics and its bankruptcy time. The equilibrium is summarized by a nonlinear valuation equation linking the fund balance, outstanding pension liabilities, annuity demand, and solvency. The framework thus turns one possible explanation of the annuitization puzzle into an equilibrium source of pension-system fragility.

**Related literature.** This paper contributes first to the literature on annuity demand. The benchmark result is that households facing uncertain lifetimes value annuities because they insure consumption against longevity risk (Davidoff et al., 2005; Yaari, 1965). Observed behavior falls well short of this benchmark. Using U.S. Health and Retirement Study data, Brown (2001) finds that the life-cycle value of an annuity predicts whether defined-contribution balances are annuitized, but leaves much of the observed choice unexplained. Dushi and Webb (2004) likewise document very low purchases of additional annuities among older U.S. households. Evidence from the United Kingdom also shows limited participation in the voluntary annuity market (Inkmann, Lopes, & Michaelides, 2011). Even when an annuity is offered directly within a pension plan, a substantial minority takes cash: in U.S. pension data, Hurd and Panis (2006) find that 20 percent of benefits offering a lump-sum option are cashed out. Taken together, these studies establish a persistent gap between the large theoretical gains from longevity insurance and retirees' willingness to annuitize, although its magnitude varies with the institutional setting (Benartzi et al., 2011).

The literature offers several explanations for this gap. Adverse selection raises the price and lowers the money's worth of voluntary annuities because individuals with private information that they are long-lived are more likely to purchase them (Finkelstein & Poterba, 2002). Fixed nominal annuities expose retirees to inflation risk, while inflation-indexed products have historically been scarce in private markets (Brown, Mitchell, & Poterba, 2002). Public pensions and defined-benefit plans also pre-annuitize a large share of household wealth, reducing the marginal value of purchasing additional longevity insurance (Dushi & Webb, 2004). Health and medical-expense shocks create a separate demand for liquid wealth. When high medical costs are correlated with changes in mortality, the remaining value of an annuity can be especially low precisely when those expenses arise (Reichling & Smetters, 2015). Bequest motives, other liquidity needs, and the framing of annuity products

provide additional explanations (Brown et al., 2008).

Most closely related to our mechanism, Li, Neumuller, and Rothschild (2021) show that real and perceived provider-insolvency risk, together with imperfect information about guaranty protection, can substantially reduce annuity demand and welfare. In their analysis, households take the provider’s insolvency risk as given when choosing how much to annuitize. In our setting, insolvency is endogenous: retirees’ choices between lump-sum and monthly benefits alter the fund’s cash flows and hence its probability of depletion. Annuity demand, provider solvency, and the value of the annuity are therefore jointly determined in equilibrium.

Second, the paper relates to research on the sustainability and reform of public pensions. Demographic aging erodes the contribution base and expands benefit obligations, creating adjustments in taxes, benefits, retirement ages, labor supply, and private saving (Attanasio, Kitao, & Violante, 2007; Börsch-Supan, Härtl, & Leite, 2016). Structural analyses quantify how pension institutions and reforms reshape retirement and life-cycle decisions (Daminato & Padula, 2024; Delalibera, Ferreira, & Machado Parente, 2026; French & Jones, 2011; Kitao, 2018). Our focus is complementary. We hold the principal policy parameters fixed and ask how a benefit-withdrawal option can propagate financial stress even before a reform occurs. This introduces beneficiary composition—the division of new retirees between immediate and deferred claims—as an endogenous determinant of pension solvency.

Relatedly, our mechanism connects to evidence that retirement decisions respond to expectations about future pension policy. Bi and Zubairy (2023) show that the timing of pension retrenchments matters: immediately implemented reforms delay retirement, whereas reforms announced with long implementation lags can bring retirement forward, particularly when the reform is fundamental or trust in government is low. We likewise emphasize forward-looking responses to the credibility of future benefits. In our setting, however, the relevant uncertainty arises endogenously from the fund’s financial state, and retirees’ responses feed back into that state rather than merely responding to an externally specified reform path.

Finally, the paper is related to models of runs and maturity choice. The classic bank-run literature shows how individually optimal withdrawals can generate aggregate fragility (Diamond & Dybvig, 1983; Goldstein & Pauzner, 2005). The closest structural analogue is the dynamic debt-run model of He and Xiong (2012), in which creditors arrive sequentially and decide whether to roll over maturing short-term debt. We retain the stochastic fundamental, staggered decisions, and endogenous default time, but the underlying claim is different. A retiree chooses once between an immediate payment and a life-contingent stream; this choice changes both current outflows and the evolving stock of monthly-benefit recipients. The pension run therefore operates through endogenous benefit maturity and demographic turnover, rather than repeated rollover of fixed-maturity debt.

## 2. Institutional Background

### 2.1. Labor Insurance old-age benefits

The empirical setting is the old-age benefit provided by Taiwan’s Labor Insurance program. This social-insurance benefit is distinct from the Labor Pension system, under which employers contribute to workers’ individual retirement accounts. Labor Insurance is financed collectively by premiums and the assets of the Labor Insurance Fund, and benefits are administered by the Bureau of Labor Insurance (BLI).

An old-age pension was introduced on January 1, 2009. An insured worker with at least 15 years of coverage may claim a monthly pension after leaving employment and reaching the statutory claim age. The statutory age increased gradually from 60 and reaches 65 in 2026. A worker may claim up to five years early or defer for up to five years; the monthly benefit is reduced or increased by 4 percent for each year. At the normal claim age, the BLI pays the larger amount implied by

$$p = \max \{0.00775 wY + 3,000, 0.0155 wY\}, \quad (1)$$

where  $w$  is the average of the worker’s highest 60 monthly insured salaries and  $Y$  is years of coverage.

The 2009 reform created the choice studied in this paper. Workers with any Labor Insurance coverage before December 31, 2008 may choose either the monthly pension or the pre-reform *one-time old-age benefit*; the choice is irreversible once the BLI approves and pays the claim. Workers first insured in 2009 or later cannot choose the pre-reform benefit. The one-time benefit generally pays one month of average insured salary for each of the first 15 coverage years and two months for each additional year, subject to a ceiling of 45 months before age 60 and 50 months including post-60 coverage. This grandfathered payment is the lump sum in our model. It should not be confused with the separately defined *old-age lump-sum benefit* for workers with fewer than 15 years of coverage, whom we exclude from the choice sample.

The statutory formulas make the annuity potentially much more generous than the lump sum. For example, a worker with an average insured salary of NT\$45,800 and 30 years of coverage receives NT\$2.061 million under the one-time formula but NT\$21,297 per month under the second formula in [equation \(1\)](#). If the worker receives the pension for 12 years, its present value at a 2 percent annual discount rate is approximately NT\$2.72 million. The monthly option therefore offers valuable longevity insurance as well as a larger expected transfer, but it also creates a persistent liability for the common fund. This difference is the institutional basis for the tradeoff in our model.

Taiwan’s government has made direct budgetary contributions to the Labor Insurance Fund since 2020. Accordingly, “bankruptcy” in the model does not mean that current law

mechanically terminates benefits when recorded assets reach zero. It denotes exhaustion of the fund under existing contribution and benefit rules—the point at which continued payment requires an additional fiscal transfer or a reform. The baseline estimation ends in December 2019 to identify the fund’s endogenous dynamics before regular government injections; the later observations remain useful for describing choices and validating the entry reconstruction.

### 3. Data

#### 3.1. Sources and sample construction

We construct a monthly panel beginning in January 2009, when the old-age pension was introduced. The current data vintage ends in June 2026, the latest month for which all component series are available. The panel combines three official sources. First, monthly reports from the Bureau of Labor Funds (BLF) provide the exact month-end balance and investment return of the Labor Insurance Fund. We read the balance from the report’s asset-allocation table rather than its rounded narrative total. The fund included both ordinary and occupational-accident insurance through April 2022; following the institutional separation in May 2022, the reported Labor Insurance Fund contains only ordinary insurance. Second, BLI statistical tables provide the stock of old-age-pension recipients, pension recipients receiving their first payment, and claims for the one-time old-age benefit. Third, Ministry of the Interior (MOI) population files provide monthly registered population and annual population and deaths by age.

We retain only one-time old-age benefit claims as lump-sum choices. In particular, we exclude the old-age lump-sum benefit for short coverage histories and the old-age difference payment because neither represents the same choice between the grandfathered lump sum and the monthly pension. Let  $L_t$  denote the number of qualifying one-time claims in month  $t$ ,  $Q_t$  the stock of monthly pension recipients, and  $A_t$  the number entering the monthly scheme. The empirical monthly-choice share is

$$q_t = \frac{A_t}{A_t + L_t}. \quad (2)$$

The two state variables used in estimation are the fund balance per registered resident,  $B_t/N_t$ , and monthly pension recipients per resident,  $Q_t/N_t$ .

#### 3.2. Reconstructing entries into the monthly scheme

The BLI reports  $Q_t$  throughout the sample, but identifies first-payment recipients only from May 2022 onward. A change in  $Q_t$  is not itself an entry measure because recipients leave the stock when they die. For the earlier period, we reconstruct monthly entries from the

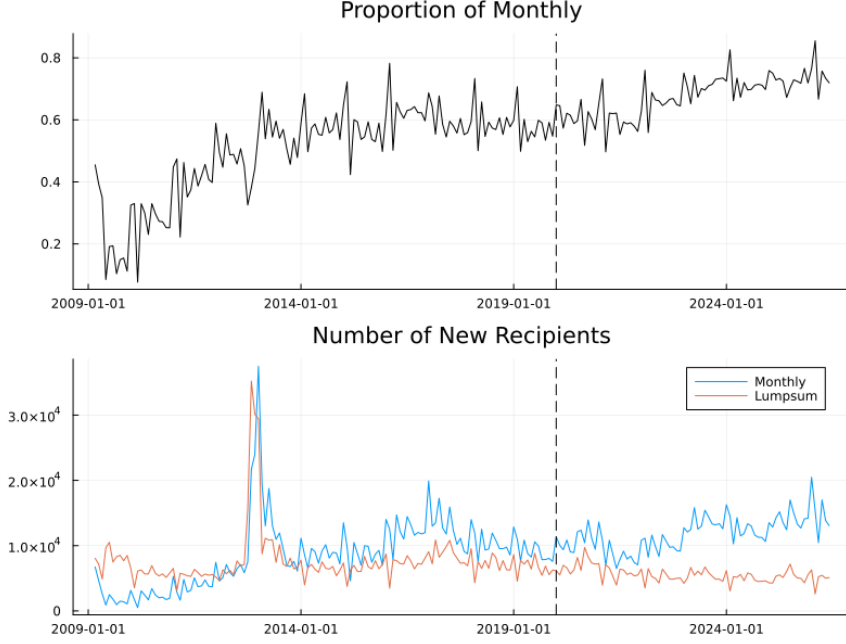


Figure 1: New Labor Insurance old-age benefit recipients

accounting identity

$$\hat{A}_t = \max \{0, Q_t - Q_{t-1} + \delta_t Q_{t-1} + c\}, \quad (3)$$

where  $\delta_t$  is the monthly mortality probability for Taiwan residents aged 60 and older. We calculate the corresponding annual death probability from MOI age-specific deaths and population, then convert it to a monthly probability as  $1 - (1 - d_t)^{1/12}$ . If the current year's age table has not yet been released, we carry forward the latest available annual rate.

The constant  $c$  corrects for systematic differences between the stock-flow identity and reported first payments. We estimate it over the post-May-2022 overlap as the mean of  $A_t - (Q_t - Q_{t-1} + \delta_t Q_{t-1})$ . In the current data vintage, the overlap contains 50 months and yields  $c = -2,452.84$  recipients per month; the mean absolute reconstruction error in that validation period is 179 recipients. We use reported first payments whenever they are available and the adjusted reconstruction beforehand. Equation (3) assumes that mortality is the only exit from the recipient stock and applies the mortality of all residents aged 60 and older to Labor Insurance beneficiaries. The reconstructed early series should therefore be interpreted as a measurement of annuity entry rather than an exact administrative count.

## 4. Model

We consider a continuous time model with an infinite horizon. There is a continuum of individuals choosing between the lumpsum and the monthly pension schemes. The pension fund is running by the government with the return and the volatility being exogenously given. When the pension system goes bankrupt, the recipients cannot receive any more

pensions from then on.

## 4.1. Population

Let  $b(s), m(s)$  be the fertility rate and the mortality rate respectively. Let  $g(t, s)$  be the unnormalized population density. The population dynamic is given by

$$\frac{\partial g}{\partial t} = - \underbrace{\frac{\partial g}{\partial s}}_{\text{aging}} - \underbrace{mg}_{\text{mortality}} + \underbrace{\int bgds\delta_0}_{\text{birth}} \quad (4)$$

where  $\delta_0$  denotes the Dirac delta centered at 0. To simplify the model, we take the assumptions that  $b(s) = b$  is a constant and that  $m(s) = m\mathbf{1}\{s \geq T_m\}$ . In plain words, this means that only those who have aged  $T_m$  exposes to the mortality risks.<sup>1</sup>

Let  $N_t = \int_0^\infty g(t, s)ds$  be the total population and  $\hat{g}(t, s) = \frac{g(t, s)}{N_t}$  be the normalized population density. Assuming that  $\hat{g}$  is at its steady state, an analytical solution is available:

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m \\ be^{-ns}e^{-m(s-T_m)} & s \geq T_m \end{cases} \quad (5)$$

where  $n = \frac{\dot{N}_t}{N_t}$  is the population growth rate satisfying

$$1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}. \quad (6)$$

For a detailed derivation, we refer to [appendix A](#).

## 4.2. Pension System

We assume that all households retire at age  $T$ . When retiring, the households choose between a lumpsum transfer  $\ell$  and a monthly pension that pays a stream of income  $p$ . Let  $Q(t, s)$  be the population density among the monthly pension recipients. The dynamic is given by

$$\frac{\partial Q}{\partial t} = - \underbrace{\frac{\partial Q}{\partial s}}_{\text{aging}} - \underbrace{mQ}_{\text{mortality}} + \underbrace{q_t \hat{g}(T) N_t \delta_T}_{\text{new retirees}} \quad (7)$$

where  $q_t$  is the proportion of new retirees choosing pension. Suppose that  $T \geq T_m$ . We can further simplify the recipients dynamic thanks to our assumption on the mortality rate. Let

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<sup>1</sup>We recognize that this is a somewhat restrictive and unrealistic assumption, but as we are going to see, this assumption helps to greatly simplify the computation.

$Q_t = \int_T^\infty Q(t, s)ds$ .  $Q_t$  follows the law of motion

$$dQ_t = [q_t \hat{g}(T)N_t - mQ_t] dt. \quad (8)$$

Let  $B_t$  be the level of the pension fund. Its law of motion is given by

$$dB_t = \left[ \underbrace{rB_t}_{\text{interest}} + \underbrace{\tau y N_t \int_{T_w}^T \hat{g}(s)ds}_{\text{pension tax}} - \underbrace{(1 - q_t)\ell N_t \hat{g}(T)}_{\text{lumpsum}} - \underbrace{pQ_t}_{\text{monthly}} \right] dt + \sigma B_t dZ_t, \quad (9)$$

where  $Z_t$  is a Brownian motion.  $\tau$  is the tax rate;  $y$  is the income ;  $T_w$  is the age when individuals start to work.

The system goes bankrupt when  $B_t$  hits 0. Denote the remaining time for  $B_t$  to hit 0 by  $T_B$ . We shall consider the detrended states  $\hat{B}_t = \frac{B_t}{N_t}$ ,  $\hat{Q}_t = \frac{Q_t}{N_t}$ . Since  $N_t$  is positive in finite time, the bankrupt time is equivalent to the time when  $\hat{B}_t = 0$ . In other words, the the remaining time for  $\hat{B}_t$  hitting 0,  $T_{\hat{B}}$ , coincides with  $T_B$ . The law of motions for the transformed states are given by

$$d\hat{B}_t = \left[ (r - n)\hat{B}_t + \tau y \int_{T_w}^T \hat{g}(s)ds - (1 - q_t)\hat{g}(T)\ell - p\hat{Q}_t \right] dt + \sigma \hat{B}_t dZ_t \quad (10)$$

$$d\hat{Q}_t = [q_t \hat{g}(T) - (m + n)\hat{Q}_t] dt. \quad (11)$$

To simplify the notation, we also denote the drift terms by  $\mu_B$  and  $\mu_Q$ , respectively.

### 4.3. Individuals' Pension Decisions

A retiring individual  $i$  chooses between between the two schemes  $d \in \{\ell, p\}$  to maximize the utility, given by

$$u_{it}(d) = \begin{cases} \alpha + \log(\ell) + \epsilon_i(\ell) & \text{if } d = \ell, \\ \log(V_t) + \epsilon_i(p) & \text{if } d = p, \end{cases} \quad (12)$$

where  $V_t$  is the present discount value of the monthly pension given the current states  $(\hat{B}_t, \hat{Q}_t)$ ,

$$V_t = V(\hat{B}_t, \hat{Q}_t) = \mathbb{E} \left[ \int_0^{T_{\hat{B}}} e^{-(\rho+m)s} p ds \mid \hat{B}_t, \hat{Q}_t \right] \quad (13)$$

with  $\rho$  being the subjective discount factor. We assume that the  $\epsilon_i(d)$  are independent and identically distributed across individuals and the choices, following the standard Gumbel distribution. The shift parameter  $\alpha$  controls the preference of the lumpsum scheme over the monthly one. Although we do not model here elaborately, this may be thought to represent the need of liquidity or the incomplete pension market. Another parameter  $\beta$  controls the sensitivity to the flucutation in the present discount value of the monthly pension.

By the property of the Gumbel distribution, the individuals' choices aggregate to the proportion choosing the monthly scheme

$$q_t = \mathbb{P}_t(u_{it}(p) > u_{it}(\ell)) = \frac{1}{1 + \exp(\alpha + \log(\ell) - \log(V_t))} \quad (14)$$

by the law of large number across the individuals. Note that  $q_t$  is a function of  $\hat{B}_t$  and  $\hat{Q}_t$  since  $V_t = V(\hat{B}_t, \hat{Q}_t)$ .

#### 4.4. Equilibrium

We now list down the equilibrium conditions. Given the Markovian states  $(\hat{B}_t, \hat{Q}_t)$ , the equilibrium is determined by the present discount value of the monthly pension  $V_t$  and the proportion choosing monthly scheme  $q_t$  such that

- (a) Given  $q_t$ , the law of motions of the states  $(\hat{B}_t, \hat{Q}_t)$  is consistent with  $q_t$  (equation (10)).
- (b) Given the law of motions of the states, the present discount value of the monthly pension  $V(\hat{B}_t, \hat{Q}_t)$  is consistent with the law of motions (equation (13)).
- (c) The individuals optimize and the decisions aggregate to  $q_t$  (equation (14)).

To determine the equilibrium, it suffices to compute  $V(\hat{B}, \hat{Q})$ . Notice that  $V$  satisfies the following Hamilton-Jacobi-Bellman equation

$$(\rho + m)V = p + \mathcal{A}V, \quad (15)$$

with the boundary condition  $V(0, \hat{Q}) = 0$  for all  $\hat{Q}$ . The operator

$$\mathcal{A}f = \mu_B \frac{\partial f}{\partial B} + \mu_Q \frac{\partial f}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2 f}{\partial B^2} \quad (16)$$

is the infinitesimal generator. Since the drift terms also depend on  $V$ , this partial differential equation is nonlinear.

To solve the model numerically, we impose the artificial boundary at the upper bound of  $\hat{B}$ . Since clearly  $V_t$  has upper bound  $\frac{p}{\rho+m}$  and is increasing in  $\hat{B}$ , it follows that  $\frac{\partial V}{\partial B}$  is approximately 0 for  $\hat{B}$  sufficiently large.

We solve the model based on the finite difference method alongside the successive iteration that shares the same spirit with Howard's policy iteration method. This numerical scheme can be viewed as a semi-implicit scheme with time step size being infinite. The sparsity of the generator matrix is exploited to achieve a better efficiency. We present our algorithm explicitly in algorithm 1.

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**Algorithm 1** Successive Iteration

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**Require:** Initial guess  $V^{(0)}$ , tolerance  $\epsilon > 0$ , maximum iterations  $K$

1: Set  $k \leftarrow 0$

2: **repeat**

3:     Given  $V^{(k)}$ , compute

$$\mu_B^{(k)} = \mu_B(B, Q, V^{(k)}(B, Q)), \quad \mu_Q^{(k)} = \mu_Q(B, Q, V^{(k)}(B, Q))$$

and the infinitesimal generator

$$\mathcal{A}^{(k)} = \mu_B^{(k)} \frac{\partial}{\partial B} + \mu_Q^{(k)} \frac{\partial}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2}{\partial B^2}.$$

with adjustments to the boundary nodes.

4:     Solve the linear system

$$\left( (\rho + m)I - \mathcal{A}^{(k)} \right) V^{(k+1)} = \mathbf{b}, \quad \mathbf{b} = p \operatorname{vec} \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & 1 \end{pmatrix}.$$

5:     Compute the error

$$e^{(k+1)} \leftarrow \left\| V^{(k+1)} - V^{(k)} \right\|_{\infty}.$$

6:      $k \leftarrow k + 1$

7: **until**  $e^{(k)} < \epsilon$  or  $k = K$

8: **return**  $V^{(k)}$ .

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## 5. Results

### 5.1. Calibration

We separate the parameters in the model in to two parts. One is calibrated outside the model and the other is estimated by solving the inverse problem. To illustrate the estimation procedure, denote the estimand vector  $\theta$ . The data is a time series  $X_i = (\hat{B}_i, \hat{Q}_i)$  sampled from equidistant time grids of gap size  $\Delta$ .

The parameters for the population structure are calibrated to Taiwan's population. We take  $T_m = T_r = 65$  and  $T_w = 20$ . The birth rate is set to be the total fertility rate in 2025 and the mortality rate leads to the median lifespan being 77. The details are given in [table 1](#).

Table 1: Model Parameters

| Parameter                            | Description                    | Value   | Target                                                                                                                    |
|--------------------------------------|--------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| <i>Panel A: External Calibration</i> |                                |         |                                                                                                                           |
| $b$                                  | Birth rate                     | 0.00462 | Total fertility rate                                                                                                      |
| $m$                                  | Mortality rate                 | 0.06    |                                                                                                                           |
| $T_w$                                | Working age                    | 25      |                                                                                                                           |
| $T_r$                                | Retirement age                 | 61      | Average retirement age                                                                                                    |
| $T_m$                                | Age of mortality exposure      | 60      |                                                                                                                           |
| $r$                                  | Pension fund return rate       | 0.057   | Mean annualized pension fund return                                                                                       |
| $\tau$                               | Pension tax                    | 0.075   | Pension tax                                                                                                               |
| $y$                                  | Income                         | 5.268   | Highest monthly insured salary                                                                                            |
| $\ell$                               | Lump-sum pension               | 19.75   | Transfer based on 30 years of working                                                                                     |
| $p$                                  | Monthly pension                | 2.449   | Pension based on 30 years of working                                                                                      |
| <i>Panel B: Internal Calibration</i> |                                |         |                                                                                                                           |
| $\sigma$                             | Pension fund volatility        | 0.1002  | $SD(\frac{\hat{B}_{t+\Delta} - \hat{B}_t - \mu_B(\hat{B}_t, \hat{Q}_t, \theta)\Delta}{\hat{B}_t \sqrt{\Delta}}) = \sigma$ |
| $\alpha$                             | Preference for lump-sum scheme | -1.205  | $E[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2]$                                                                              |
| $\rho$                               | Subjective discount rate       | -0.0599 | $E[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2]$                                                                              |

Note: The monetary unit in the model is 100,000 NTD.

## Appendix

### A. Derivation of the Population Equations

To find the steady state  $\hat{g}$ , conjecture that  $g(t, s) = \hat{g}(s)N_t$ . Plugging back to the population equation gives

$$\hat{g}(s) \frac{\dot{N}_t}{N_t} = -\frac{\partial \hat{g}}{\partial s} - m(s)\hat{g}(s) + b\delta_0. \quad (17)$$

Notice that the right hand side does not depend on  $t$ . Hence  $\frac{\dot{N}_t}{N_t} = n$  is a constant. We can also write the Dirac delta as part of the boundary condition:

$$\hat{g}(s)n = -\frac{\partial \hat{g}}{\partial s} - m(s)\hat{g}(s), \quad \hat{g}(0) = b. \quad (18)$$

This is a standard initial value problem with  $m(s) = m$  for  $s \geq T_m$  and 0 otherwise. The solution is given by

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m \\ be^{-ns}e^{-m(s-T_m)} & s \geq T_m. \end{cases} \quad (19)$$

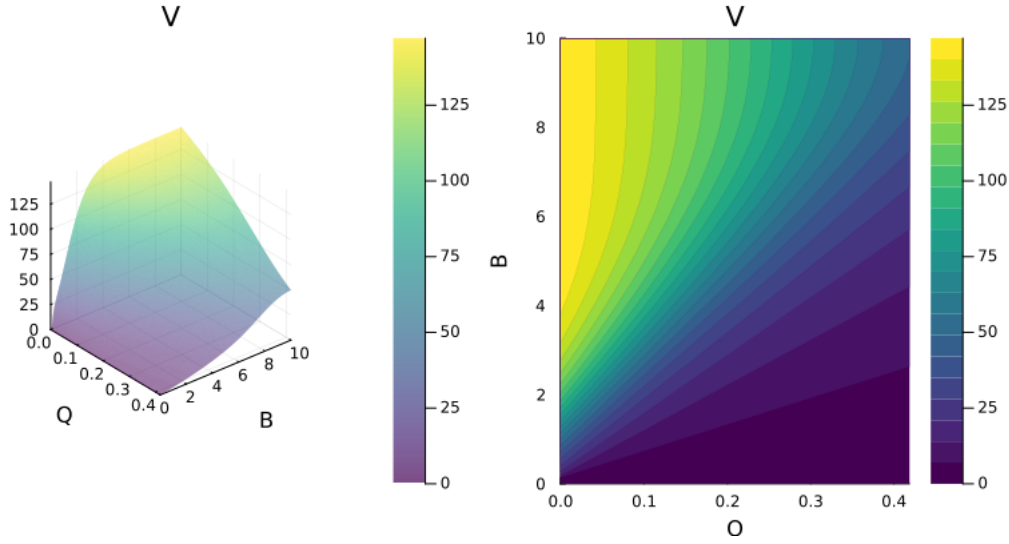


Figure 2: Estimated  $V$

Finally, the population growth rate  $n$  can be determined by the normalizing condition

$$\int_0^{\infty} \hat{g}(s) ds = 1 \quad \Rightarrow \quad 1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}. \quad (20)$$

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